

ACA Marketplace Instability: A Predictive Framework for Community-Level Healthcare Access Risk

DATA 404: Advanced Python for AI & Data Science
Howard University

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B. Abstract

This Data 404 project explores how we can predict healthcare access risk in the Affordable Care Act (ACA) Marketplace by looking at more than just insurance data. When insurers leave a marketplace, the effects impact real people through higher costs, fewer choices, and disruptions in care.

To better understand these risks, we use a multimodal approach that brings together insurer participation data (CMS), demographic and socioeconomic indicators (ACS), policy variables (KFF), and economic conditions (BLS). Instead of treating these inputs as isolated features, we organize them into distinct modalities and incorporate interaction terms to capture how these factors influence one another.

We implemented an early fusion modeling strategy. The modalities are integrated into a unified feature space and used to train classification models, including Logistic Regression, Random Forest, and Gradient Boosting. Results show how models that capture interactions across modalities perform better at identifying high-risk states.

C. Introduction

The ACA Marketplace was designed to make healthcare more accessible. However, its stability depends on the insurance decision to remain in the market. When insurers leave, fewer plans, higher premiums, and fewer options remain for families.

What makes this problem more complicated is that insurer exits do not happen in isolation. They are influenced by factors such as local economic conditions, state policies, and the characteristics of the population they serve. Right now, government

reporting systems look at marketplace stability after changes occur. What is missing is the ability to analyze issues at a community level and see risk building ahead of time.

As outlined in the proposal, current approaches do not fully account for how different types of data interact to signal instability. This project focuses on:

- Predicting where insurer exits are more likely to happen
- Understanding how economic, demographic, and policy factors work together
- Comparing models to see which ones handle this complexity best
- Translating results into visuals that are easy to interpret

D. Data Description

Instead of relying on one dataset, we combined multiple sources to capture different aspects of marketplace behavior:

- Centers for Medicare & Medicaid Services (CMS): insurer participation, plan counts, enrollment
- American Community Survey (ACS): income, poverty, uninsured rates
- Kaiser Family Foundation (KFF): policy decisions like Medicaid expansion
- Bureau of Labor Statistics (BLS): unemployment and economic stress

E. Methods

Rather than treating everything as just numbers in one table, we think about the data as different modalities:

- Market behavior (CMS)

- Population vulnerability (ACS)
- Policy environment (KFF)
- Economic pressure (BLS)
- Time trends (year-over-year changes)

Data Preparation

- Missing values were handled and cleaned
- Data was standardized across states and years
- All datasets were merged into a state-year structure

We tested three main models:

- Logistic Regression
- Random Forest
- Gradient Boosting

We introduced interaction terms that capture how these different modalities influence one another. For example, economic stress combined with low insurer participation can increase the risk of marketplace instability. This tells us that healthcare access is not shaped by one factor alone. Access is based on overlapping conditions that disproportionately impact marginalized and underserved communities. This helps us to identify communities that may be vulnerable before disruptions occur and provide data to support more proactive policy decisions.

We designed the process to reflect how different modalities interact:

- Features were grouped based on data type (economic, policy, demographic, market)

- We engineered interaction features (e.g., economic stress, insurer participation)
- Time-based features were added to capture trend behavior

To make sure our results were reliable, we used a cross-validation approach. The data was split into five parts and the model was trained and tested multiple times across different combinations. For the classification models, we used stratified splits so that each fold maintained a balanced representation of the target classes. In addition to validating the models, we also tuned key hyperparameters to improve performance. We adjusted the number of trees in ensemble models and modified learning rates to control how the model learns.

F. Results

When we compared model performance, Logistic Regression provided a reliable baseline. As we introduced more flexible models, we noticed some improvements. The Random Forest model increased predictive accuracy by capturing difficult relationships in the data. Gradient Boosting performed the best and delivered the highest level of predictive performance.

For example, Logistic Regression achieved a ROC-AUC of approximately 0.85, compared to about 0.74 for a simpler Decision Tree model. In healthcare access, multiple factors work together at the same time. And models that are able to account for nonlinear patterns and interactions between variables tend to better reflect the challenges of real-world problems.

The dashboard visualizations show how risk appears across different dimensions. The risk map provides a geographic view of how marketplace instability is distributed across the U.S. The “Top 10 High-Risk States” visualization highlights where instability is most likely to occur. Risk trends over time show how risks can change from year to year, and whether conditions are improving or worsening. Lastly, the feature-versus-risk plot illustrates how certain factors (e.g., market share) are associated with increased risk.

G. Explainability & Interpretation

To understand what drives marketplace instability, market concentration, unemployment rates, the differences in state-level policies help explain where and why risk is more likely to occur. Instability tends to emerge when multiple factors overlap. For example, communities with high economic stress might also experience limited insurer competition. When combined with certain policy environments, these conditions can increase the risk of disruption.

H. Discussion

A major strength of this project is its ability to bring together different types of data to understand the challenges of the healthcare system better. By combining economic, demographic, policy, and market data, the model captures patterns that would be missed if each dataset were analyzed in isolation. Plus, the use of clear visualizations helps translate technical results into observations that are easier to understand and communicate.

However, there are important limitations to consider. Using state-level data means that local differences within communities may not be fully captured, and the time range of the data may limit the ability to identify longer-term trends. Some policy variables are also simplified. It is important to recognize that socioeconomic data can reflect deeper structural inequalities and model predictions should be interpreted with care. The goal is not to label communities as “high risk,” but to use the analysis to better support them.

From an ethical perspective, it is important to remember that the goal is to predict outcomes and to prevent harm before it happens. These models should be used as tools for better decision-making, and not just to label certain communities as “high risk.” If used responsibly, these observations can help policymakers act earlier, give communities time to prepare for disruptions, and enable healthcare systems to respond more effectively.

I. Conclusion

This project tells a story of how bringing together different types of data can strengthen our ability to predict healthcare access risk. Rather than reacting to marketplace instability after it happens, this approach focuses on identifying early warning signs so that action can be taken sooner. At its core, this work is about people. By using data to better understand where and why disruptions may occur, we can help support communities that are often the most affected by changes in healthcare access.

As students at an HBCU, this perspective is especially important to us. Many of the communities we come from and serve are directly impacted by these systems, and this project reflects a commitment to using data science for advocacy and impact.

Looking ahead, there are several ways this work can continue to grow.

Future improvements could include incorporating county-level data to capture local conditions better, or integrating real-time updates to make predictions more responsive. There is also an opportunity to expand the modeling approach by exploring more advanced multimodal architectures that process different types of data separately before combining them. This would allow for a deeper understanding of how multiple factors interact over time. The ultimate goal is to build tools that predict risk and help create a better healthcare system by informing decisions that protect and support underserved communities.

Appendix A: Data Dictionary: The following table shows key variables used in the multimodal dataset. Variables are grouped by data source and reflect economic, demographic, policy, and market-based inputs used to model healthcare access risk.

Variable Name	Description	Data Source	Type	Example
state	U.S. state identifier	CMS	Categorical	GA
year	Year of observation	CMS	Numeric	2018
num_insurers	Number of insurers participating in marketplace	CMS	Numeric	5
plan_count	Total number of available plans	CMS	Numeric	32
enrollment	Total ACA marketplace enrollment	CMS	Numeric	245,000
market_concentration	Measure of insurer dominance in market	Derived (CMS)	Numeric	0.65
poverty_rate	% of population below poverty line	ACS	Numeric	18.2
median_income	Median household income	ACS	Numeric	52,300
uninsured_rate	% of population without insurance	ACS	Numeric	12.5
population_size	Total state population	ACS	Numeric	10,500,000
unemployment_rate	State unemployment rate	BLS	Numeric	6.1
labor_force_participation	% of working-age population employed	BLS	Numeric	62.4
medicaid_expansion	Medicaid expansion status (0 = No, 1 = Yes)	KFF	Binary	1

state_marketplace	Type of marketplace (state/federal)	KFF	Categorical	Federal
policy_index	Composite score of policy environment	Derived (KFF)	Numeric	0.72
insurer_exit	Indicator of insurer exiting market (target variable)	CMS	Binary	1
year_over_year_change	Change in insurer participation from previous year	Derived	Numeric	-2
econ_market_interaction	Interaction: unemployment × number of insurers	Engineered	Numeric	30.5
policy_demo_interaction	Interaction: policy × poverty rate	Engineered	Numeric	12.7
risk_score	Predicted probability of marketplace instability	Model Output	Numeric	0.78

Appendix B: Model Performance Summary: The following table compares model performance across classification models used to predict insurer exit risk.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.82	0.79	0.76	0.77	0.85
Decision Tree	0.75	0.72	0.70	0.71	0.74
Random Forest	0.86	0.84	0.81	0.82	0.89
Gradient Boosting	0.88	0.86	0.84	0.85	0.91

Appendix C: Hyperparameter Configuration: The following table outlines key hyperparameters used during model training and tuning.

Model	Parameter	Value
Logistic Regression	max_iter	1000
Decision Tree	max_depth	10
Random Forest	n_estimators	100
Random Forest	max_depth	10
Gradient Boosting	learning_rate	0.05
Gradient Boosting	n_estimators	100

Appendix D: GitHub Repository

The full project is available in the GitHub repository below. This repository serves as the central location for collaboration, version control, and reproducibility of results.

GitHub Link:

<https://github.com/kayode-ddv/DATA404-Group-Project>

The repository includes:

- Data ingestion and cleaning scripts
- Feature engineering and dataset construction

- Model training and evaluation code
- Visualization files and outputs
- Documentation of the project workflow

Team Contributions

This project was a collaborative effort. Each team member contributed across different areas of the workflow. Kayode Dunkel Vance focused on data processing and engineering; collecting, cleaning, and merging datasets from multiple sources. Camryn Davis handled the model, including feature engineering, training machine learning models, and evaluating performance.

Gabriele McKinley was responsible for the data visualization portion. She created dashboards and visual outputs that helped translate our model results. Lastly, Parker Jessie focused on storytelling, explaining the technical analysis piece, and the writing of the final report. Each team member contributed to data preparation, model discussion, visualization review, and overall project development.

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